SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Cryptography and Network Security**
COURSE CODE: **CS A5113**

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Ch.Nagasaisrikanth/192311159 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q. No	*Understanding of Concepts (25)*	*Experimental Design (30)*	*Use of Tools (20)*	*Analysis of Results (15)*	*Documentation (10)*	*Total Score (out of 100)*
1	5 /5	5 /6	4 /4	3 /3	2 /2	19
2	5 /5	5 /6	4 /4	3 /3	2 /2	19
3	5 /5	5 /6	4 /4	3 /3	2 /2	19
4	5 /5	5 /6	4 /4	3 /3	2 /2	19
5	5 /5	5 /6	4 /4	3 /3	2 /2	19
OVERALL RUBRICS VALUE						
	/5	/5	/4	/4	/2	95
	25 /25	25 /30	20 /20	15 /15	10 /10	/100

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
   a. Show differences and similarities
      i. Map how each affects the Data structure and Security level
      ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
   a. Show relationships between these concepts
   b. Analyze how each contributes to secure communication
   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CO-1 ASSESSMENT TOOL -2

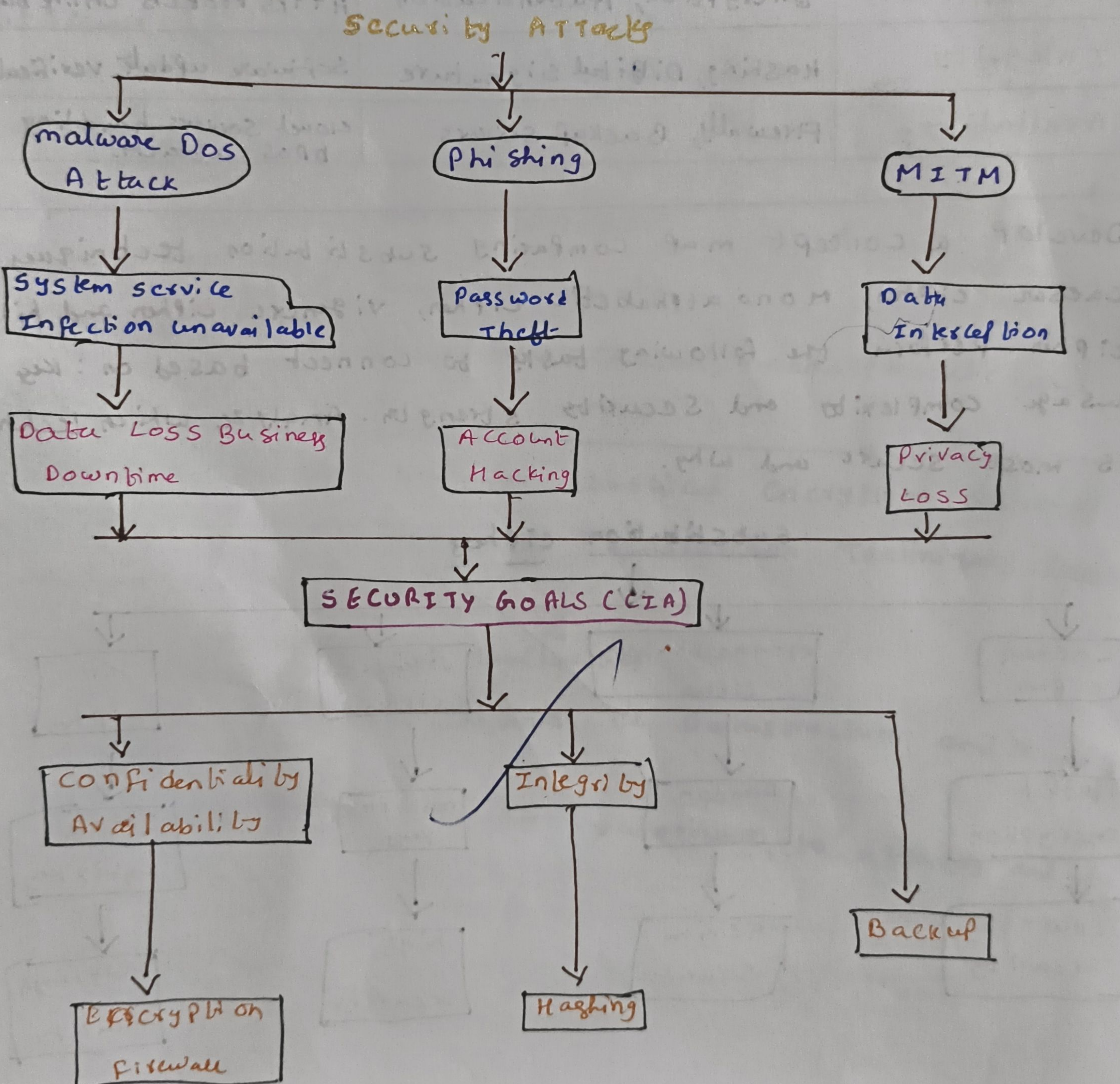
CONCEPT MAP

Ch.Nagasai Srikanth
192311159

Create a concept map linking the following security Attacks Threats, Risks and security Goals (confidentiality, Integrity Availability, Perform the following tasks.

a) show how each attack leads to specific threats and risks

b) Map how security goals help mitigate them provide one real world example for each link.



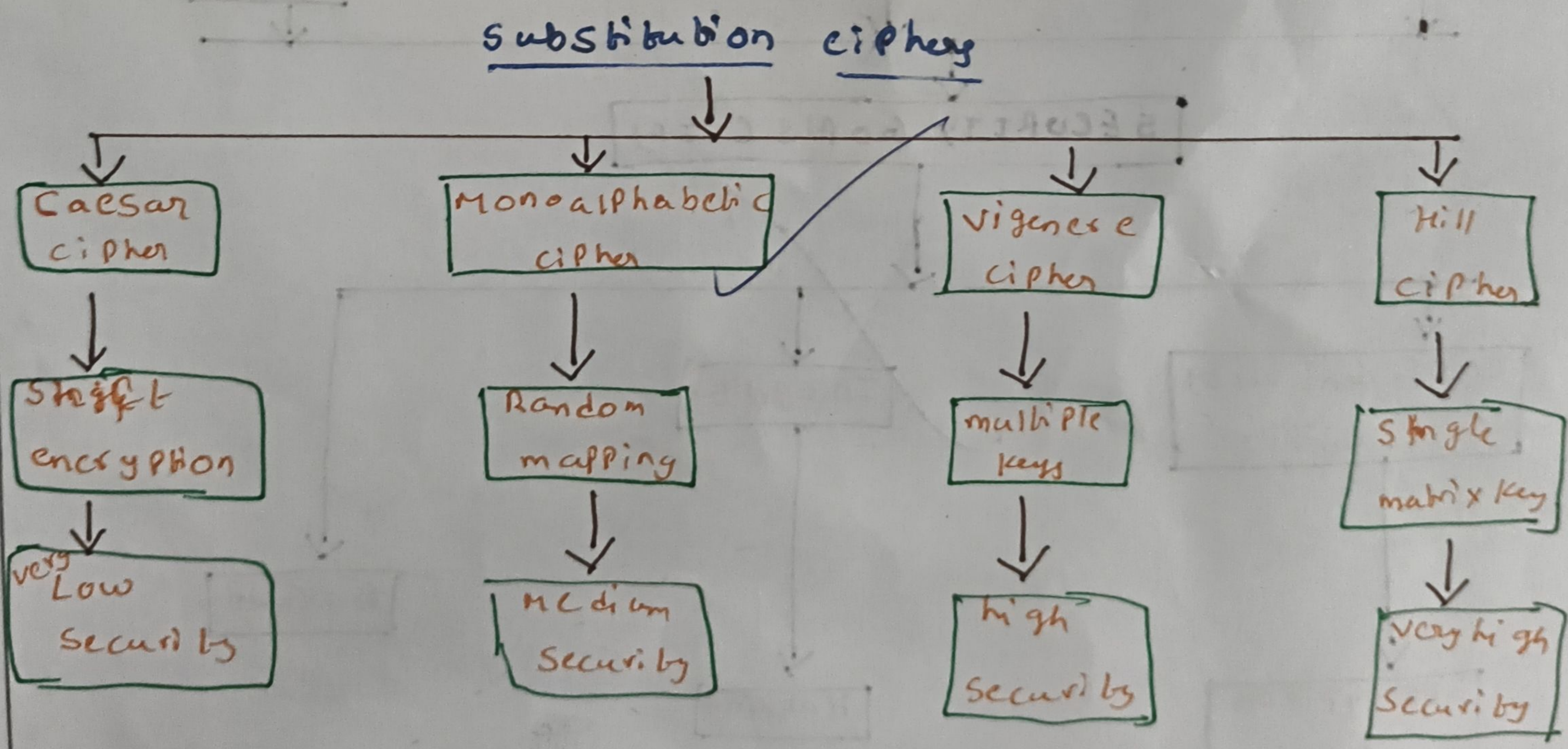
a) Attack → threat → Risk

Attack	Threat	Risk
Malware	System infection	Data loss
Phishing	Password theft	Account compromise
Man in the Middle	Data interception	Information leakage
Denial of service	Service disruption	website downtime.

b) Security Goal & Real world example

Security Goal	Mitigation	Real-world Example
Confidentiality	Encryption, Authentication	HTTPS Protects online banking
Integrity	Hashing, Digital signature	Software update verification
Availability	Firewalls, Backup servers	cloud Servers handling DDOS attacks.

2) Develop a concept map comparing substitution techniques caesar cipher, Monoalphabetic cipher, vigenere cipher and hill cipher. perform the following tasks to connect based on: key usage complexity and security strength. Analyze which technique is most secure and why.



Cipher	Key usage	Complexity	Security Strength
Caesar	single shift	very low	very weak
monoalphabetic	Random alphabet	medium	moderate
vigenere	keyword	High	strong
Hill cipher	matrix key	very high	strongest.

Analysis:

Most secure: Hill cipher

Reason:

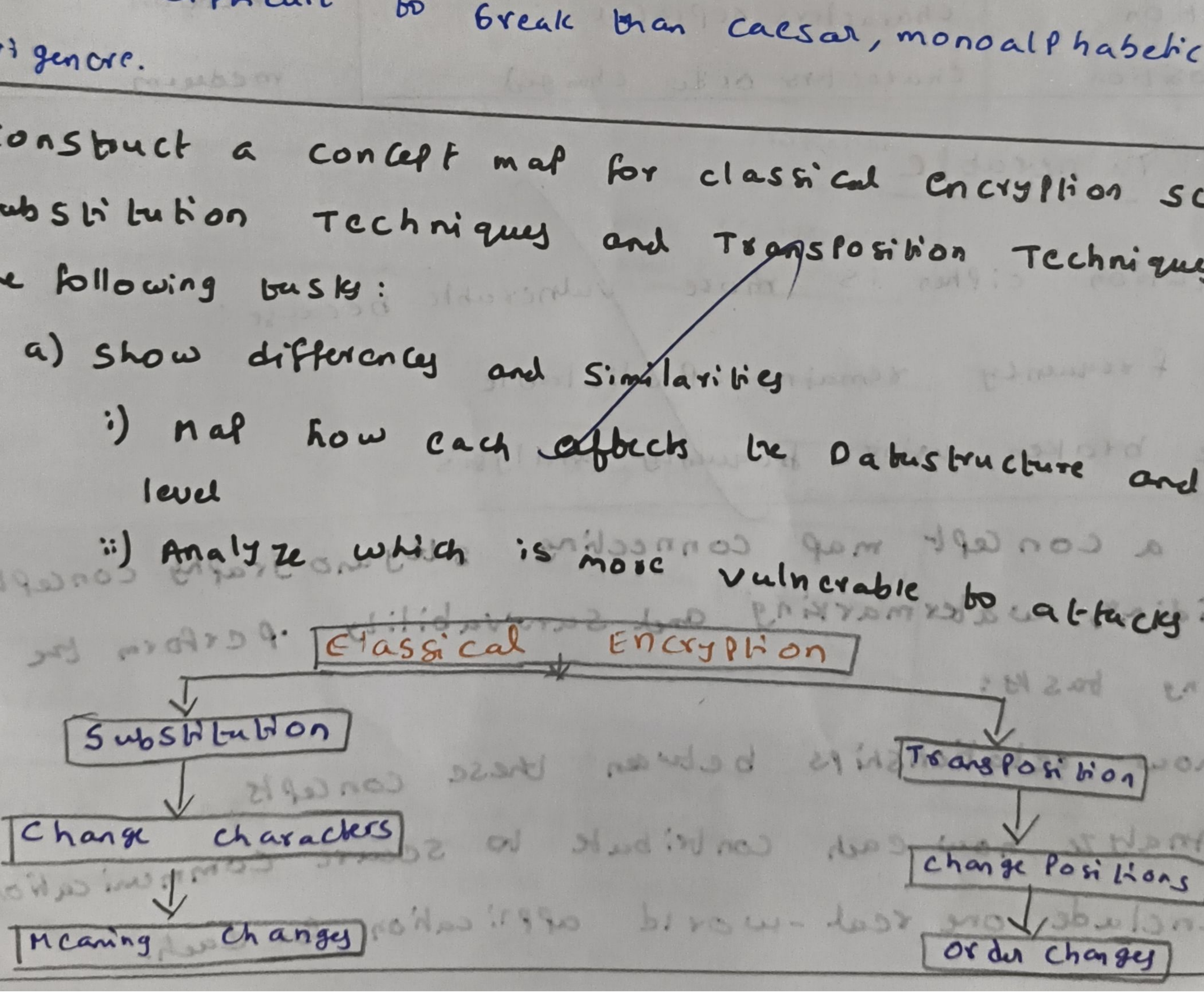
- uses matrix multiplication
- Encrypts multiple letters together.
- Resistant to simple frequency analysis
- More difficult to break than caesar, monoalphabetic and vigenere.

3) Construct a concept map for classical encryption schemes: substitution Techniques and Transposition Techniques. Perform the following tasks:

a) show differences and similarities

i) map how each affects the Datastructure and security level

ii) Analyze which is more vulnerable to attacks and why.



a) Similarity

-> Both are classical encryption methods

-> Both use secret keys

-> Both Protect Plaintext

Differences

Substitution	Transposition
Replaces letters	Rearranges letters
character changes	Position changes
Frequency changes	Frequency remains same.

i) Effect on Data Structure and security

Technique	Data Structure	Security level
Substitution	characters replaced	medium
Transposition	Characters order changed	medium

ii) More vulnerable

Substitution cipher is more vulnerable because :

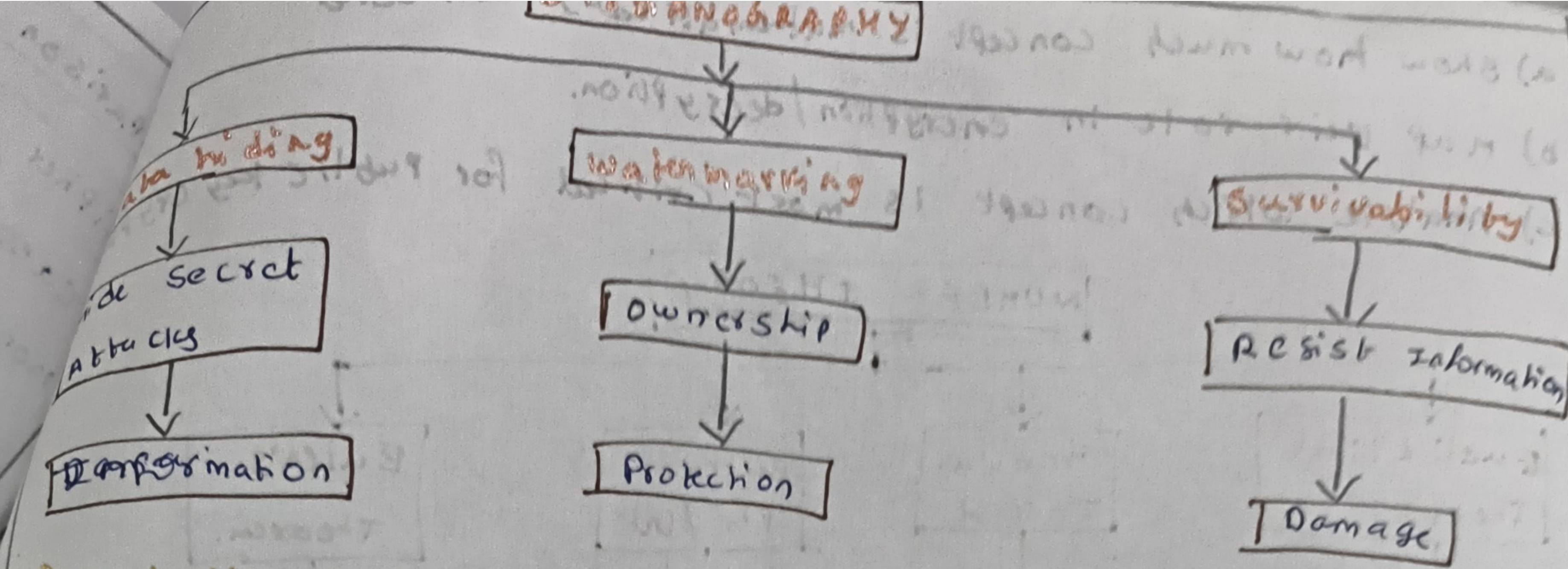
- Letters frequency remains predictable
- can be broken using frequency analysis

4) Create a concept map connecting steganography concepts, Data hiding, watermarking and survivability. Perform the following tasks:

a) show relationships between these concepts

b) Analyze how each contribute to secure communication

c) Include one real-world application for each.



a) Relationship

-> Data hiding hides confidential information
-> watermarking proves ownership
-> Survivability ensures hidden data remains intact after compression

b) contribution to secure communication

Concept	contribution
Data hiding	secret communication
watermarking	copyright protection
Survivability	Reliable recovery of hidden data.

c) Real world Applications

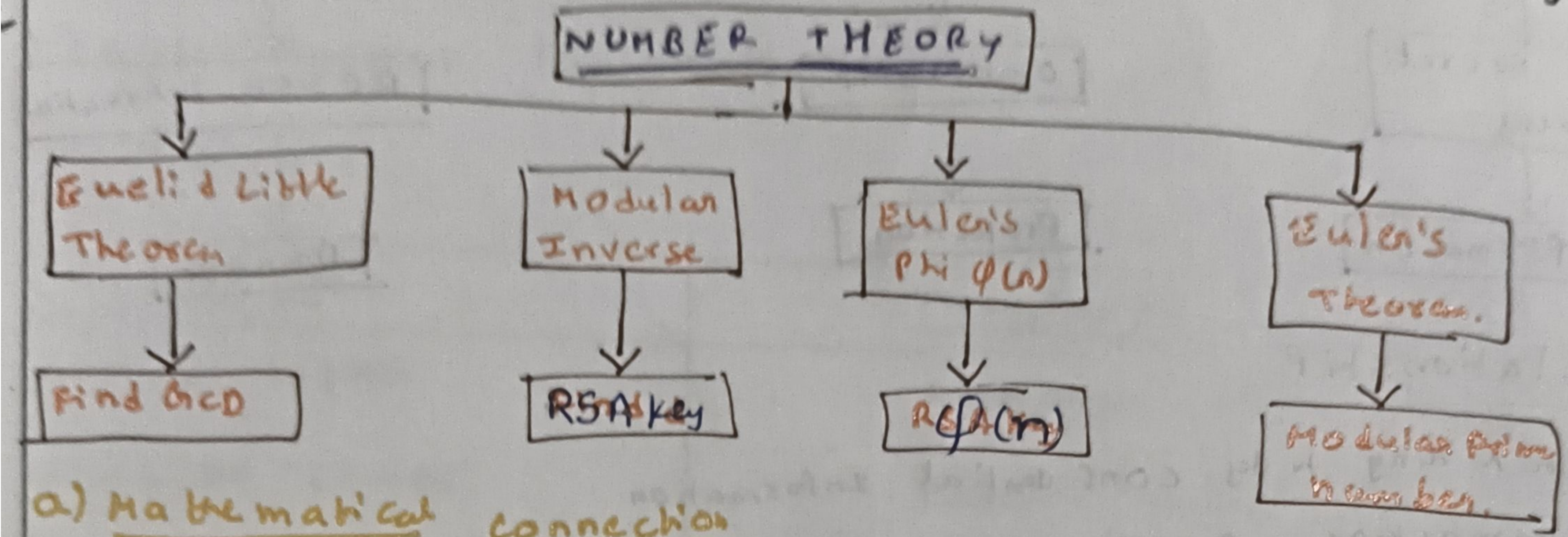
concept	Example
Data hiding	secret militiary communication
watermarking	copyright protection for digital Images
survivability	medical Image authentication

5) Draw a concept map linking number theory concepts used in cryptography. Modular Inverse. Extended Euclid Algorithm. Fermat's little theorem, Euler's Phi Function and Euler's theorem.

a) show how much concept is mathematically connected

b) map their role in encryption/decryption.

c) Analyze which concept is most critical for Public key cryptography



a) Mathematical connection

* Extended Euclidean Algorithm → Find GCD and modular Inverse
* Modular Inverse → used for RSA Private key
* Euler's phi function → calculates $\varphi(n)$.
* Euler's theorem → Basis of RSA encryption / decryption.

b) Role in Encryption/Decryption

Concept	Role
Extended Euclidean Algorithm	computes modular Inverse
Modular Inverse	Generates RSA Private key
Euler's Phi function	calculates RSA keys
Euler's theorem	Enables RSA encryption / decryption.

c) most critical concept.

Euler's Theorem is the most critical because:

- It forms the mathematical foundation of the RSA algorithm
- Ensures that encrypted messages can be decrypted correctly using the corresponding Private key.